Higher National Diploma in Civil Engineering

Fourth Year, 1st Semester Examination – 2018

CE 3222 - Foundation Engineering

Instructions for candidates:
Answer any five questions.
Calculators are allowed.

No. of Pages : 04
No of questions: 06

Time: Three (03) hours

Q1.

(a) Briefly explain the meanings of Ultimate bearing Capacity and Allowable bearing Capacity.

(04 marks)

(b) State three types of shear failures in soil and briefly explain them (06 marks)

(c) Determine the size of square footing at a depth of 1m below ground to carry net allowable load of 295kN as shown in figure 1. Assume a factor of safety of 3 and using Terzaghi equation for general shear failure. Assume that the water table is at the ground surface level.

(10 marks)

Q2.

(a) The cross section of the cantilever retaining wall is shown in figure – 02. Calculate the factor of safety with respect to overturning. Neglect the passive pressure and Ka = 0.35.

(10 marks)

(b) State two checks other than overturning check to be performed for the stability of the retaining wall. (04 marks)

(c) State three types of retaining walls (03 marks)

(d) Indicate the three main parts of a retaining wall in a sketch. (03 marks)

Q3. (a) Define the terms void ratio and porosity with the help of a phase diagram. (04 marks)

(b) Derive an expression for void ratio 'e' in terms of porosity 'n' using the phase diagram

(04 marks)

(d) A sample of saturated clay has a mass of 1,853.5 g, and 1,267.4 g after drying. The dry unit weight is 14.71 kN/m3. Find the following:

(i) Bulk density γ_{wet};
(ii) Void ratio e;
(iii) Specific gravity of soil Gs;
(iv) Bulk density γ_{wet} for Degree of saturation S = 50 percent

(12 marks)

Q4.

(a) State any four situations where deep foundations are required? (04 marks)

(b) A pile is driven through soft cohesive deposit overlying stiff clay; the average un-drained shear strength in the soft clay is 50 kPa and in the lower deposit the average un-drained shear strength is 150 kPa. The water table is 5 m below the ground level and the stiff clay is at 8 m depth. The unit weights are17.5 kN/m^3 and 19 kN/m^3 for the soft and the stiff clay respectively. **Estimate the length** of 500 mm diameter pile to carry a load of 500 kN with a safety factor of 4, using α - method (figure Q4)

(10 marks)

(c) Draw the vertical effective pressure on pile with depth (06 marks)

Q5.

(a) If you have been appointed for the foundation design of a ten storied building what would be your first activity to carry out that design of foundation?

(2marks)

(b) State four details that you would expect from the above mentioned activity

(4 marks)

(c) Illustrate the foundation types that you are familiar with.

(6 marks)

(d) Briefly explain three difficult conditions of soil.

(3 marks)

(e) If this building is proposed to be constructed on such a difficult condition of soil, clearly explain with reasons the types of foundation that are suitable?

(5 marks)

Q6. (a) State three types of caissons. (3 marks)

(b) Briefly explain the situations where caisons are suitable. (5 marks)

(c) A open circular caisson of 5m internal diameter and 7m external diameter is driven into soft clay. The depth of the caisson into the ground is 18m and it is located 15m below the water table. The unit weight of the concrete is 23.55kN/m^3. f_c' is 28 days characteristic strength is 21x10^3 kN/m^2 undrained cohesion of the soil is 30kN/m^2.

(i) Determine the thickness of bottom plug.
(ii) Do check for perimeter shear.
(iii) Do check for buoyancy

(12marks)

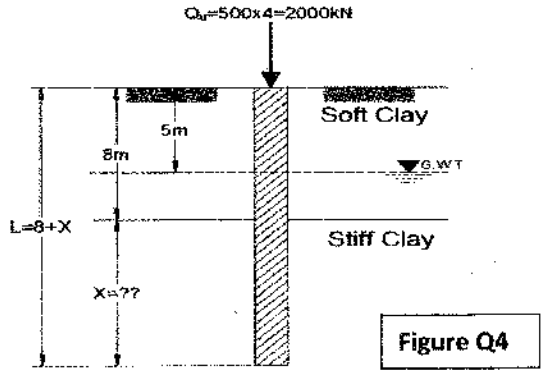


Figure Q4

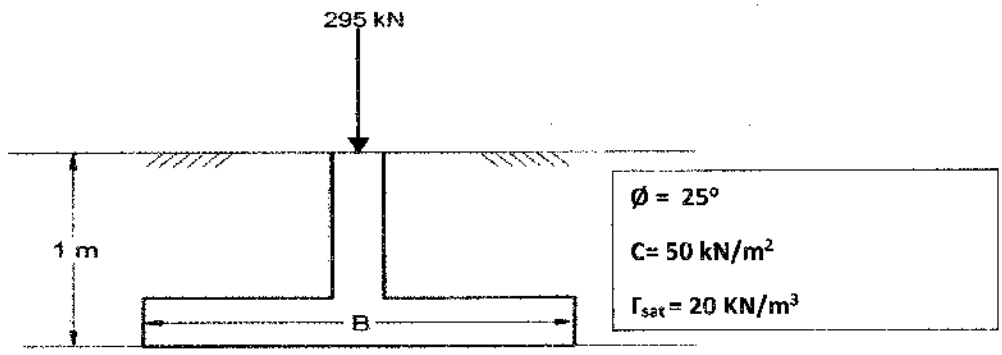
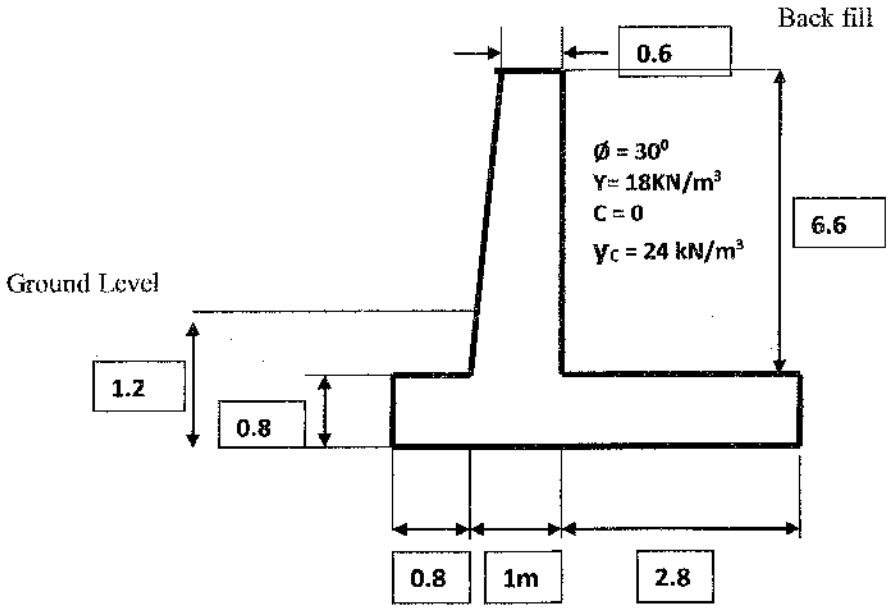


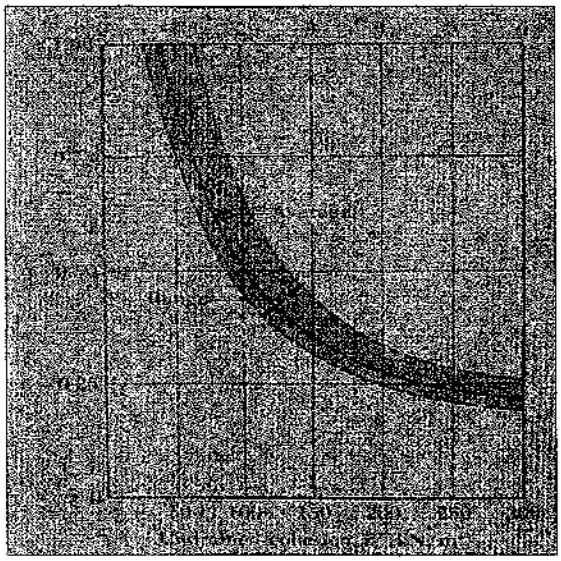
Figure Q1



All dimensions in meter

Figure Q2

$V_U = 0.85t\sqrt{f_c}$



Ø degrees	Nc	Nq	Nγ
0	5.7	1.0	0.0
5	7.3	1.6	0.14
10	9.6	2.7	1.2
15	12.9	4.4	1.8
20	17.7	7.4	5.0
25	25.1	12.7	9.7
30	37.2	22.5	19.7
35	57.8	41.4	42.4
40	95.7	81.3	100.4
45	172.3	173.3	360.0
50	347.5	415.1	1072.8

$$t = 1.18R_i\sqrt{\frac{q}{f_c}} \qquad \text{(circular caisson)}$$

t – Thickness of plug, Ri – Internal radius, q – unit bearing pressure, f_c – Concrete flexural strength (0.1 – 0.2 of fc', fc' 28 days compressive strength of concrete

$$q \approx H\gamma_w - t\gamma_c$$